

When the World Changed: The Limits of FPAS Mark I

Susanna Grigoryan

Recent Staff Member, Monetary Policy Department, Central Bank of Armenia

Student, Global Forecasting School

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Note

1. FPAS Mark I and Its Strengths

FPAS Mark I frameworks, inspired by the work of Freedman and Laxton (2009), were built on a clear and simple logic. They combined macroeconomic models, expert judgment, and structured policy processes to ensure internal consistency between forecasts and decisions. The approach worked effectively in relatively stable times. Baseline forecasts provided a coherent narrative about where the economy was heading, and policy recommendations could be evaluated against this most-likely outlook. For two decades, this approach supported credible inflation-targeting regimes across advanced and emerging- market economies.

2. The Three Layers of Uncertainty

The world that FPAS Mark I was designed for assumed a stable structure and modest, mean- reverting shocks. But beginning in the late 2010s, three forms of uncertainty began to dominate policymaking:

- Structural uncertainty - permanent changes to the global economy caused by COVID-related shocks, geopolitical realignments, and deglobalization.
- Model uncertainty - the erosion of stable relationships such as the Phillips Curve, with nonlinearities, supply constraints, and credibility effects.
- Policy-credibility uncertainty - the risk that inflation expectations could de-anchor when shocks persist and communication loses clarity.

These layers of uncertainty made it impossible to rely on a single, deterministic baseline as the guide for decisions. The very foundation of FPAS Mark I-the idea that the central bank could identify and optimize around one “most-likely” scenario-became untenable.

3. The Limits of Baseline Determinism

In the classical FPAS structure, the baseline scenario represented the best estimate of the future. Policy recommendations were derived from this single narrative. But the experience of 2020–2023 demonstrated how fragile such determinism can be. When multiple shocks interact—pandemics, supply bottlenecks, energy crises—there is no single trajectory the economy will follow. Instead, central banks face a range of possible outcomes, many of them shaped by behavioural feedback, credibility shifts, and nonlinear responses.

A deterministic baseline not only misrepresents reality; it also biases communication. By projecting a false sense of precision, it discourages policymakers from discussing risks openly. As shown in *The Prudent Risk Management Approach to Price Stability* (Central Bank of Armenia, 2025), effective communication under uncertainty requires transparency about risks—not just point forecasts.

4. Lessons from 2020–2023

The pandemic and subsequent global supply shocks, as well as geopolitical tensions, exposed the operational limits of FPAS Mark I. Across both advanced and emerging economies, baseline-driven forecasts failed to anticipate the persistence of inflationary pressures. In some cases, this was due to structural changes in labour markets and global trade; in others, it reflected insufficient integration between judgment and model-based analysis. Central banks that clung to deterministic baselines were repeatedly forced to revise forecasts upward and backtrack on forward guidance. Confidence in the forecasting process—and by extension, in monetary policy itself—suffered.

The experience reaffirmed that policy under uncertainty cannot be based on a single optimal path. Instead, it must recognize and manage the spectrum of possible risks that define the new environment.

5. Implications – Toward FPAS Mark II

The recognition that uncertainty is not a temporary disturbance, but a structural condition led to a profound rethinking of the FPAS architecture. The Central Bank of Armenia, drawing on lessons from global experience, developed the *Prudent Risk Management Approach to Price Stability*—FPAS Mark II. This new system embeds risk analysis within every stage of the policy process, from forecasting to communication. Its guiding principle is prudence: minimizing potential regret rather than optimizing a single baseline scenario. By embracing uncertainty, FPAS Mark II transforms it from a source of vulnerability into a foundation for robust policymaking.

Literature & Further Reading

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Script

Hello everyone, I’m **Susanna Grigoryan**, an economist at the **Central Bank of Armenia** and a **Level One student at the Global Forecasting School**. In this video, we’ll explore how the world changed—and why traditional forecasting systems could no longer keep up.

The evolution of modern monetary policy frameworks has always reflected the world around them. During the 1990s and early 2000s, the first generation of Forecasting and Policy Analysis Systems—FPAS Mark I—served central banks remarkably well. They provided structure, discipline, and coherence in policy formulation and communication. But by the early 2020s, the global environment had changed beyond recognition. Pandemic disruptions, geopolitical conflicts, and deglobalization revealed deep vulnerabilities in these deterministic systems. The experience underscored a profound truth: the world had become too uncertain for single-baseline policymaking.

FPAS Mark I frameworks were built on a clear and simple logic. They combined macroeconomic models, expert judgment, and structured policy processes to ensure internal consistency between forecasts and decisions. The approach worked effectively in relatively stable times. Baseline forecasts provided a coherent narrative about where the economy could be heading, and policy recommendations could be evaluated against

this most-likely outlook. For two decades, this approach supported credible inflation-targeting regimes across advanced and emerging-market economies.

The world that FPAS Mark I was designed for assumed a stable structure and modest, mean-reverting shocks. But beginning in the late 2010s, three forms of uncertainty began to dominate policymaking: structural uncertainty, meaning permanent changes to the global economy caused by pandemics, geopolitical realignments, and deglobalization; model uncertainty, meaning the erosion of relatively stable relationships such as the Phillips Curve, with nonlinearities, supply constraints, and credibility effects; and policy-credibility uncertainty, meaning the risk that inflation expectations could de-anchor when shocks persist and communication loses clarity. These layers of uncertainty made it impossible to rely on a single, deterministic baseline as the guide for decisions. The very foundation of FPAS Mark I—the idea that the central bank could identify, optimize and communicate around one “most-likely” scenario—became untenable.

In the classical FPAS structure, the baseline scenario represented the central bank’s best and “most plausible” estimate of the future, and policy recommendations were derived from this single narrative. But the experience of 2020 to 2023 demonstrated how fragile such determinism can be. When multiple shocks interact—pandemics, supply bottlenecks, energy crises, wars—there is no single trajectory the economy will follow and it is impossible to derive a most “plausible scenario.” Instead, central banks face a range of possible outcomes, many of them shaped by behavioral feedbacks, credibility shifts, and nonlinear responses. A single deterministic baseline not only misrepresents reality; it also biases communication. By projecting a false sense of precision, it discourages policymakers from discussing risks openly. As shown in *The Prudent Risk Management Approach to Price Stability* (Central Bank of Armenia, 2025), effective communication under uncertainty requires transparency about risks, not just point baseline forecasts.

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The recognition that uncertainty is not a temporary disturbance, but a structural condition led to a profound rethinking of the FPAS architecture. The **Central Bank of Armenia**, drawing on lessons from global experience, developed the **Prudent Risk Management Approach to Price Stability—FPAS Mark II**. This new system embeds risk analysis within every stage of the policy process, from forecasting to communication. Its guiding principle is prudence: minimizing potential regret rather than optimizing a single scenario. By embracing uncertainty, FPAS Mark II transforms it from a source of vulnerability into a foundation for robust policymaking.

For many years, FPAS Mark I served central banks well. It offered structure and discipline, helping us explain our policy decisions through a single baseline forecast. But the world we live in today is not the same as it was twenty years ago. The pandemic, global supply shocks, and geopolitical tensions showed us that uncertainty is no longer the exception—it's the rule.

FPAS Mark I was built on the idea that we could predict the most likely path for the economy and optimize policy around it. That worked relatively well when the world was stable. But once multiple shocks hit simultaneously, the idea of a single baseline became unrealistic. We needed a system that could manage uncertainty, not ignore it. That's why the Central Bank of Armenia developed the **Prudent Risk Management Approach to Price Stability**, or FPAS Mark II. It's a framework designed for a world of uncertainty, where policy is not about optimizing a single baseline forecast but about managing risks and minimizing regret. By embracing this mindset, policymaking institutions can make policy decisions that remain credible, flexible, and forward-looking—no matter how unpredictable the world becomes.

Thank you for watching, and I would like to acknowledge **Douglas Laxton, Jared Laxton, Asya Kostanyan, and Sopio Mkervolidze** at the **Global Forecasting School** for their guidance and support in producing this video.

See you there.